SECTION - 12

ROLLING SHUTTERS

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SECTION – 12

ROLLING SHUTTERS

1.00 SCOPE

This specification covers the general requirements for mild steel plain / grille rolling shutter and aluminium grille rolling shutter with motor drives and/or manual operation and/or Gear operation including design, supply, fabrication and installation in accordance with the drawings and specifications.

2.00 GENERAL REQUIREMENTS

2.01 The Contractor shall furnish all materials, labour, operations, equipment, tools & plant, scaffolding and incidentals necessary and required for the completion of all metal work in connection with steel rolling shutters as called for in the drawings, specifications and Schedule of probable itemswhich cover the major requirements only. Anything called for in the tender documents shall be considered as applicable to the items of work concerned. The supply and installation of additional fastenings, accessory features and other items not specifically mentioned, but which are necessary to make a complete functioning installation shall form a part of this contract.

2.02 All metal work shall be free from defects, impairing strength, durability and appearance and shall be of the best quality for purposes specified made with structural properties to withstand safely strains, stresses to which they shall normally be subjected to.

2.03 All fittings shall be of high quality and as specified and as per approval of KEIPL/PM.

2.04 All electrical work shall be in strict accordance with the latest Indian Electricity Rules.

2.05 The Contractor shall strictly follow, at all stages of work, the stipulations contained in the Indian Standard Safety Code and the provisions of the Safety Rules as specified in the General Conditions of the Contract for ensuring safety of men and materials.

2.06 Any approval, instructions, permission, checking, review etc. whatsoever by the KEIPL/PM shall not relieve the Contractor of his responsibility and obligation regarding adequacy, correctness, completeness, safety, strength, workmanship etc.

3.00 CODES AND STANDARDS

3.01 All standards, specifications, acts, and codes of practice referred to herein shall be the latest editions including all applicable official amendments and revisions.

3.02 List of certain important Indian Standards, Acts and Codes applicable to this work is given below. However, the applicable standards and codes shall be as per but not limited to the list given below :

IS : 2074	Ready mixed paint, air drying, red oxide - zinc chrome, priming
IS : 2108	Blackheart malleable iron castings

IS : 6248 Specification for metal rolling shutters and rolling grills.

4.00 SHOP DRAWINGS

4.01 The Contractor shall submit shop drawings and / or samples of each type of roller doors and other items of metal work called for to the KEIPL/PM for his approval ahead of their use at site and to conform to the C.P.M. Chart. The shop drawings shall show full size sections of doors and other components, thickness of metal, details of construction, hardware as well as connection of doors and other metal work to adjacent work / supports. Samples of all joints and methods of fastening and joining shall be submitted to the KEIPL/PM for approval well in advance before commencing the work.

5.00 SAMPLES

5.01 Samples of all typical mild steel plain / grille rolling shutter and aluminium rolling shutter as called for shall be fabricated, assembled and erected or submitted to the KEIPL/PM as directed by him, for his Approval well in advance of their use at site.

6.00 MILD STEEL PLAIN / GRILLE/ PERFORATED ROLLING SHUTTER

6.01 Components & Fabrication

a) Curtain

i) For Plain Shutter / Perforated Shutter

Lath sections or slats for rolling shutter curtain shall be fabricated from 18 gauge tested bright, cold rolled annealed steel strips and machine rolled at 75 mm rolling centres with effective bridge depth of 12 mm interlocked with each other and ends locked with heavy type malleable MS/CI clips conforming to IS : 2108. The profile shall be such as to prevent excessive deflection at specified wind load. In case of peforated shutter lath section shall have 3mm perforation at 3mm clear gap.

Bottom lock plate of curtain shall be made of 3.15 mm thick (10 gauge) MS plate and 95 mm wide reinforced with suitable angle section at the bottom. The lock plate shall be interlocked with bottom most lath section of curtain so as to provide contact against the sill, when closed. When shown on drawings, a flexible weather strip shall be applied to make tight contact with the floor. The lock plate shall be provided with pulling handles and locking arrangements on both sides as shown on the drawings or as specified and / or as directed by the KEIPL/PM.

ii) For Grill Shutter

Grille curtains shall be made of 16 gauge MS links of 150 mm long with 75 mm rolling centre with the link openings of 115 x 45 mm. This link shall be assembled with 3/8" dia. hard drawn seamless tubes and the connecting sides (ie. end of tubes) should be sealed with MS clips.

Bottom lock plate of curtain shall be the same as per the Plain shutter.

b) Guide Channels

The guide channels shall be of mild steel deep channel section and of rolled, pressed or built-up (fabricated) construction of 75 mm deep, 25 mm wide, and 3.5 mm thick (minimum) with required number of fixing cleats or supports for attachment to the walls or columns by means of anchor fastner/coach screws. The guide shall be of such depth as to retain the shutter under specified wind pressure.

c) Bracket plate

The bracket plate shall be fabricated out of mild steel of 3.15 mm thickness (minimum), thicker plates may be used depending upon the height of shutter. The bracket plate shall have fitted at the centre a U shaped cast iron or mild steel clamp riveted or welded to it. Since the bracket plate carries the full load of the shutter, it should have sufficient cross-sectional area to resist the shear force and it shall be held in position rigidly by means of suitable foundation bolts.

d) **Roller**

The suspension shaft of the roller shall be made of steel pipe conforming to heavy duty of latest IS : 1161 and of sufficient diameter so as to resist deflection due to the weight of the rolling shutter. The deflection shall not exceed 5 mm per metre width.

The pipes of the suspension shaft which are clamped to the bracket shall be fitted with rotatable cast iron pulleys to which the curtain is attached. The pulleys and the pipe shaft shall be connected by means of pretensioned helical springs of suitable gauge conforming to grade 2 of IS : 4454 to counterbalance the weight of the curtain and to keep the shutter in equilibrium in any partly opened position. Grease packed ball bearings or bushings of approved make shall be provided for smooth, trouble free operation. For large size rolling shutter the roller shall be provided with double row, self-aligning ball bearings. For wide openings, the roller shall be of fabricated cage type, the roller assembly shall be designed so as to be capable of producing sufficient torque to ensure easy operation of the rolling shutter in any position. The spring tension shall be adjusted by means of suitable adjustment holes on the rim of the pulleys.

e) **Hood**

The hood cover shall be made of 20 gauge galvanised sheets, suitably reinforced with stiffener to prevent sag.

f) **Gears, Worms, etc.**

All gears, worms, etc. used in the assembly of the rolling shutters shall be machine-cut. Worm gear wheels shall be of high grade cast iron or mild steel or phosphor bronze. The worms shall be of mild steel or gun-metal or phosphor bronze.

6.02 **Manually Operating Shutters**

Manually operated shutters shall be easily operable by one person. The speed of operation shall be about 0.3 metres per second.

In general, manually operated shutters shall be push-pull types for small openings. Larger shutters shall be either chain and gear operated or crank and

gear operated. The crank handle shall be removable. All shutters shall be lockable from one or both sides as desired by the KEIPL/PM.

6.03 **Power Operating Shutters**

Power unit shall be suitable for 3 phase, 50 cycle, 400 volt AC power supply and shall be either floor or wall mounted unit. The motors shall be of sufficient capacity to move the shutter in either direction at a speed of 0.3 metres per second. In addition to the Gear motor, each standard power unit shall include a magnetic brake, reversing starter with built-in overload protection, a geared limit switch and one push button station located inside the building unless otherwise stated.

It is desirable that the bottom bar of motor operated doors shall be provided with a sensitive edge, electrically connected to stop the travel of the door on meeting an obstruction.

Operating chains shall be of tested quality, heavily galvanised and with all ends rounded to assure smooth operation and hand protection.

Reduction gears shall be of high strength grey cast iron, machine moulded from machine cut patterns.

These shall be operable from a push button station conveniently located beside the door or as shown on drawings. One emergency hand chain/crank operation shall also be provided for use in case of failure of the electrical system.

6.04 **Shop Coat**

Shutters shall be painted with one coat of red oxide or zinc chromate primer paint of approved make conforming to IS : 2074, latest edition. All surfaces shall be thoroughly cleaned of rust, grease, loose mill sealed etc. before painting shop coat. Portions which will be inaccessible after assembly of units shall be given an extra coat of paint before assembly. Where specified, doors shall be galvanized and subsequently painted with one coat of zinc chromate for adhesion of field coat.

6.05 **Storage**

The fabricated shutter shall be stacked in a careful manner and supported evenly. All precautions shall be taken to ensure that the shutters are not damaged or distorted in any manner.

6.06 **Installation**

Brackets shall be rigidly fixed to the lintel, wall or at any position as required to take the full load of the shutter. The shaft along with springs shall be fixed to the brackets. The shutter shall be laid on ground and the guide channels shall be bound with it, with ties. The shutter shall then be placed in position and the top fixed with suspension shaft. The side guides channels and cover frame shall then be fixed to the wall. Pockets shall be cut to fix the brackets, clamps, guide channels, etc. and shall be grouted with cement concrete, proportion 1:2:4, or these shall be fixed by any other method as directed by the KEIPL/PM or as specified by the Manufacturer, at the Contractor's own cost. Fixing shall be done true to plumb and level such that the operation of the shutter is easy and smooth.

6.07 Scaffolding is the responsibility of the Contractor and shall be done without causing any damage to the structure and the finishes therein.

6.08 **Painting**

Door shutter shall be enamel painted to approved finish. All painting, shall be carried out as specified under "PAINTING" or as specified in the drawings or as specified in bill of quantities.

7.00 High Speed Doors

High Speed Door with European technology complies with EN 13241-1 CE certified, vertical rolling type, used for internal and external application. Opening and closing speed of door should be 1 mtr / sec. The door components are PVC coloured and one full row clear fabric curtain, Bottom Profile, Side Guide Rails, Stiffeners, Brackets, Shaft, Automation system with advance control, accessories.

PVC Curtain- European Make: PVC curtain 0.8 mm thick. PVC fabric is strengthened by internal stiffeners. Standard PVC fabric colours are RAL 3002(Red), RAL 7042 (Grey),RAL 5002 (Blue), RAL 2004 (Orange), RAL 5012(Light Blue) and RAL 1021(Yellow). Fabric standard complies with (DIN) EN 12641-2. Optional colours available on request. Clear curtain for vision panel available.

Weight of PVC curtain	:	900 gm /sqm	
Tensile strength(warp/weft)	:	4300/ 4000 N/ 5 cm	as per DIN EN ISO 1421/v1
Tear strength (warp/weft)	:	500 / 500 N	as per DIN 53363
Fire Behaviour	:	< 100 mm/ min	as per DIN 75200 ISO 3795

Bottom Profile: Aluminum bottom profile with EPDM soft edge rubber profile at bottom. Optical wired safety edge sensor as standard. Bottom profile equipped with nylon guide runner at both side for smooth operation. Powder coated with standard off white colour.

Side Guides: Aluminum side guide sleek profile with guide PVC throughout the length at both sides of profile.Powder coated with standard off white color

Bracket Plate: Mild Steel bracket plate minimum 4.0 mm thk. Bracket plate dimensions vary as per size of door. Load of shutters transfers through brackets. Side brackets equipped with pedestal bearings for free rolling movement.

Roller Shaft: Mild Steel hollow tubular pipe. Diameter vary as per width of door. Specially designed machined shaft and flange welded at both ends of tubular pipe. PVC curtain is rolled on tubular pipe.

Hood Cover: GI sheet Hood cover of 0.8 mm thickness to cover shaft and bracket assembly. Powder coated with standard off white colour.

Automation System:

Direct Drive: Electric motor- European Make.The drives are maintenance-free alternating current/three-phase current drives with safety gear. TÜV – SÜD certified wear free lubricated worm gear with integerated safety brake. The safety brake should engage automatically and almost without a jolt to prevent dropping of door curtain. Operating Voltage 400 V, 3 phase, rated frequency 50 Z, Rated current 5.5 / 3.2 A, Rated power consumption 1.25 KW, Protection class IP 54, Inbuilt limit switch, VDE Checked. Emergency operation by Manual hand crank at motor height as standard. Inbuilt Thermostat device to prevent overheating of drive.

Control: Pulse Control- European Make, 3 way push button from inside as standard. Supply voltage 3 phase 400V,frequency 50/60 Hz, Control Voltage 24VDC, Protection class IP 65, VDE Checked. The control box is equipped with a self control of all safety devices that is activated every time there is input. Advance control with features to integerate all standard and optional accessories as mentioned below.

Standard Accessories :

One 3 way Push button standard with automation system.

Optical wireless safety edge sensor

EPDM soft edge rubber profile at bottom

Additional Operating Switch from other side of opening / Extra Push Button

Remote control with receiver

8.00 Motorised Fire rated Rolling Shutter:

Motorised Fire rated Rolling Shutter complies to BS 476 Part 22, Warrington approved. Shutter constructed of glavanised steel slats,guide rails and bottom profile, Mild steel roller shaft assembly, side brackets with bearings, Imported make automation system, hardware and accessories. Auto close by relay switch after receive of 24 Vdc fire Signal . Fire Rating : 2 hour.

Slat Curtain: High Grade Special quality galvanised steel interlocked slats as per BS standards and designed edge endlock to achieve maximum holding strength.

Bottom Plate: Structured galvanised steel L shaped section welded with angle stiffener 35 x 35 x 5 mm.

Side Guides: Specially designed box type galvanised steel side guides 3.0 mm thk surface mounted type should guide the slat curtain smoothly.

Bracket Plate: Mild Steel bracket plate minimum 3.0 mm thk. Bracket plate dimensions vary as per size of shutter. Load of shutters transfers through brackets. Motor side brackets with chain-sprocket mechanism. Non motor side bracket with bearings for free rolling movement.

Roller Shaft: Heavy duty Mild Steel hollow tubular pipe Diameter vary as per width of shutter. Specially designed machined shaft and flange welded at both

ends of tubular pipe. Slat curtain is fixed to roller shaft through supporting slat by nut bolts.

Hood Cover: GI sheet Hood cover of 0.8 mm thickness.

Automation System:

Indirect Drive: Electric motor- Imported Make. 230 V , 50 Hz, 1 phase Gear motor with thermal protector, limit switches, manual override, 3 way push button switch with key lockable. Auto closing by relay switch after receive of 24 Vdc fire signal. Thermal fusible link should be used to auto close the shutter by gravity when room temperature rises above 70℃. Thermostat device intergerated as standard in all drives.

Control: In built Basic control card with integrated limit switch and autoclosing feature.

Standard Accessories :

Thermal fusible link for actuating auto close feature when temperature reaches 70℃

One 3 way Push button standard with automation system.

Two coats of Epoxy Painting of any Non-metallic RAL Colour

9.00 Motorised Rolling Shutters

Motorised Mild Steel Single Wall Rolling Shutter, vertical rolling type, used for internal and external application. The rolling shutter components are CRC Steel/ Galvanized Steel Slats as per IS 6248, MS Bottom Plate, MS Side guides,MS Brackets, MS Shaft Assembly, all with one coat of zinc chromate red oxide primer, GI Hood Cover and Hardware. Imported Make Automation System consist of indirect drive with chain wheel mechanism, Manual hand chain for emergency operation, Inbuilt control, 3 way push button, safety and operating Accessories as optional.

Slat Curtain: CRC Steel/ Galvanized Steel slats 1.2 mm thick for shutter width more than 3.5 mtr.and 0.9 mm thick for shutter width less than 3.5 mtr. as per IS 6248. Each slat section should be continuous single piece without any joint. Interlocked slat section should have distance of 75 mm between rolling centers and slat profile should be 75 mm corrugated type.

Bottom Plate: Mild Steel bottom plate 2.5 mm thick with angle stiffener 35 x 35 x 5 mm welded to full opening width for strengthening. L-shaped bottom profile with height 120 mm interlocked with bottom most slat section. Rubber sealing at bottom as an option.

Side Guides: Mild Steel side guide 75 mm wide x 3.0 mm thick for shutter width upto 6.0 mtr. ; 100 mm wide x 3.0 mm thick for shutter width between 6.0 mtr.and 8.0 mtr.; 150 mm wide x 3.0 mm thick for shutter width above 8.0 mtr. U-shaped side guide profile for guiding the slat curtain.
Bracket Plate: Mild Steel bracket plate minimum 3.0 mm thick. Bracket plate dimensions vary as per size of shutter. Load of slat curtain should transfer through brackets. Motor side bracket with chain-sprocket mechanism, and other side bracket with bearings for smooth rolling movement.

Roller Shaft: Mild Steel hollow tubular pipe ,diameter ranging from 100 mm to 300 mm; vary as per width of shutter. Machined shaft and flange welded at respective ends of tubular pipe. Slat curtain is fixed to roller shaft through supporting slat by nut & bolts.

Hood Cover: GI sheet Hood cover 0.8 mm thick. Hexagonal shape hood to cover shaft and bracket assembly.(Optional)

Automation System:
Indirect Drive: Electric motor- Imported Make. Operating on 220 V / 380 V, 1 phase or 3 phase. Motor mounted parallel to roller shaft. Emergency operation by chain wheel drive. Motor connected to shaft assembly via sprocket and chain. Motor should equipped with safety device preventing damage in case of any accident. Inbuilt Thermostat device to prevent overheating of drive.
Control: In built Basic control card with integerated limit switch.
Standard Accessories :
Push Button
One 3 way Push button standard with automation system.
Bottom Rubber Sealing
PU Painting

10.00 **Painting**

All painting shall be carried out as specified under “PAINTING” or as specified in the drawings or as specified in the Schedule of Probable Items.

11.00 **ACCEPTANCE CRITERIA**

9.01 **Shop Inspection**

After completion the manufacture of different components of the rolling shutter, an arrangement for shop inspection by the KEIPL/PM shall be made to check the conformity with approved shop drawings.

9.02 **Field Inspection**

After installing the shutters, the Contractor shall test the performance of the shutter in the presence of the KEIPL/PM. The doors shall be smoothly operable under all ambient conditions. All control and locking devices shall give fault-free performance.

12.00 **MODE OF MEASUREMENT**

12.01 Unless noted otherwise in the schedule of probable items, the method of measurement for various items shall be generally in accordance with IS 1200 subject to the following:

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